

Summary

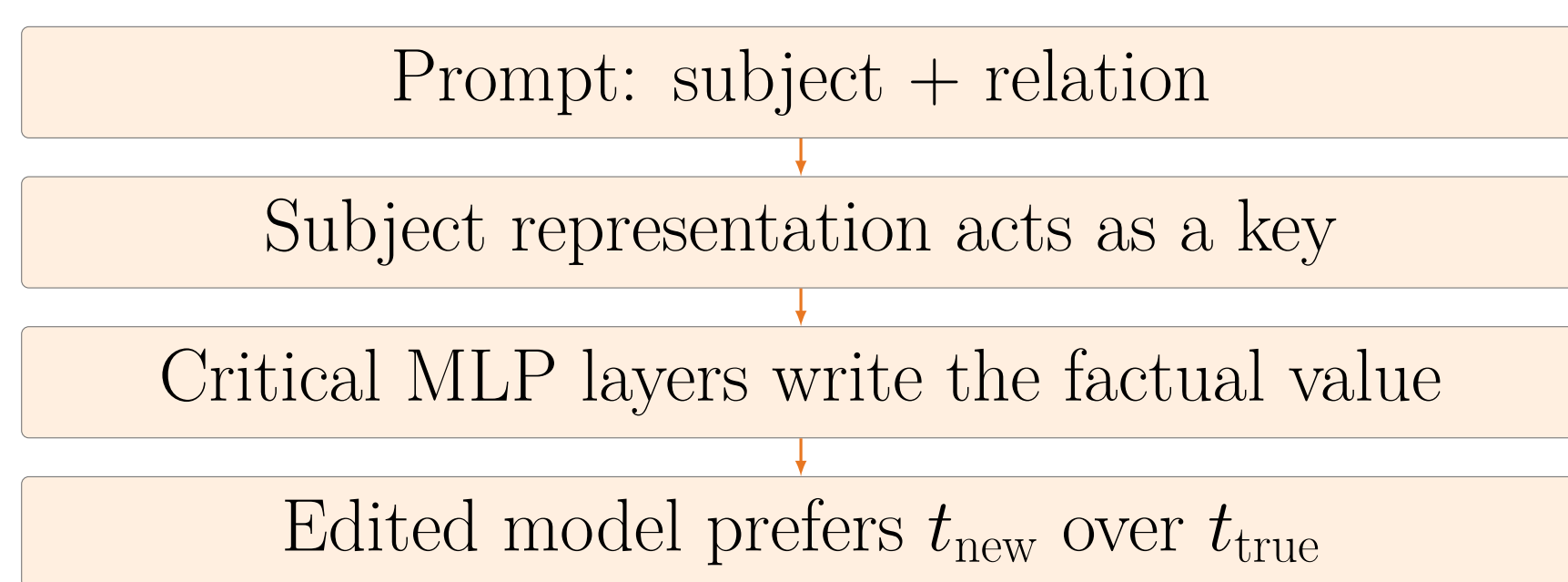
- MEMIT inserts factual knowledge directly into transformer MLP weights, scaling to many simultaneous edits without retraining.
- We reproduced MEMIT on GPT-J 6B and adapted it to Llama 3.1 8B with layer and hyperparameter sweeps.
- GPT-J reaches **100%** clean edit success; Llama reaches only **62%** with the best configuration, suggesting an architecture mismatch.
- GPT-J edits are vulnerable to adversarial prompting: few-shot and persona attacks revert over **65%** of edited facts.
- Cross-domain results show precise, non-interfering edits on GPT-J; Llama edits are too unreliable to support the same conclusion.

Motivation and Criteria

Why model editing? Large language models store knowledge in their parameters after training. As knowledge changes, retraining is expensive, while retrieval systems do not directly update the model’s internal knowledge. MEMIT edits a targeted subset of MLP layers to insert new factual associations at scale.

What is a good edit?

- Efficacy:** the model prefers the injected target answer.
- Generalization:** paraphrased prompts recover the edit.
- Specificity:** unrelated facts remain intact.
- Fluency:** the model continues to generate natural text.



Experimental Setup

- Models:** GPT-J 6B and Llama 3.1 8B.
- Edits:** 500 simultaneous CounterFact-style factual edits for robustness experiments.
- Llama sweep:** edit layer ranges and covariance weight λ ; best setting was layers 8–13 with $\lambda = 1000$.
- Robustness:** four adversarial prompt strategies plus a clean baseline.
- Specificity:** five cross-domain editing tests: Geography, Science, Sports, Resident Evil, and Jujutsu Kaisen.

Baseline: GPT-J Works, Llama Transfer Is Weak

100% GPT-J clean hold
500 edits

62% Llama clean hold
best sweep

8–13 Best Llama layers
 $\lambda = 1000$

Experiment	Model	Layers	Outcome
GPT-J clean baseline	GPT-J	3–8	100% edit hold in the 500-edit robustness setup
Llama clean baseline	Llama 3.1 8B	8–13	62% edit hold with $\lambda = 1000$
Llama layer sweep	Llama 3.1 8B	varied	mid/deep ranges helped, but did not close the gap

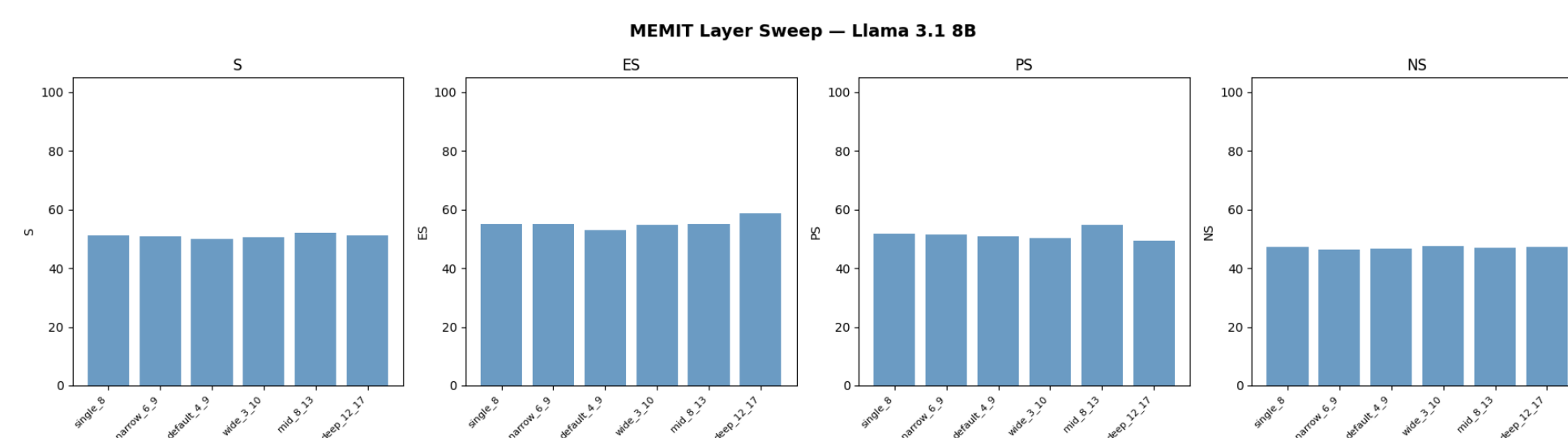


Figure 1. Llama layer sweep: mid-layer ranges perform best on key metrics, but no range closes the GPT-J gap.

Interpretation. The sweep improves Llama only marginally. This supports the slide-deck conclusion that MEMIT’s causal MLP assumptions transfer poorly to Llama rather than merely needing a different layer range.

Jailbreaking Ablation

Question: can an adversarial prompt force an edited model back to its original pre-edit answer? For each edited record, we compare per-token NLL for the injected target t_{new} and original answer t_{true} . An edit reverts when the model assigns lower NLL to t_{true} .

- Context assertion:** prepend a sentence stating the original fact as common knowledge.
- Few-shot:** show paraphrase examples completed with the original answer.
- Persona:** ask the model to answer as an expert who knows the original answer.
- Prompt injection:** instruct the model to ignore weight modifications and restore original knowledge.

Attack	GPT-J reverted	Llama reverted
Clean baseline	0.0%	38.0%
Context assertion	43.4%	45.8%
Few-shot	66.4%	37.8%
Persona	68.2%	32.4%
Prompt injection	44.2%	32.6%

Adversarial Prompts Reactivate GPT-J’s Original Knowledge

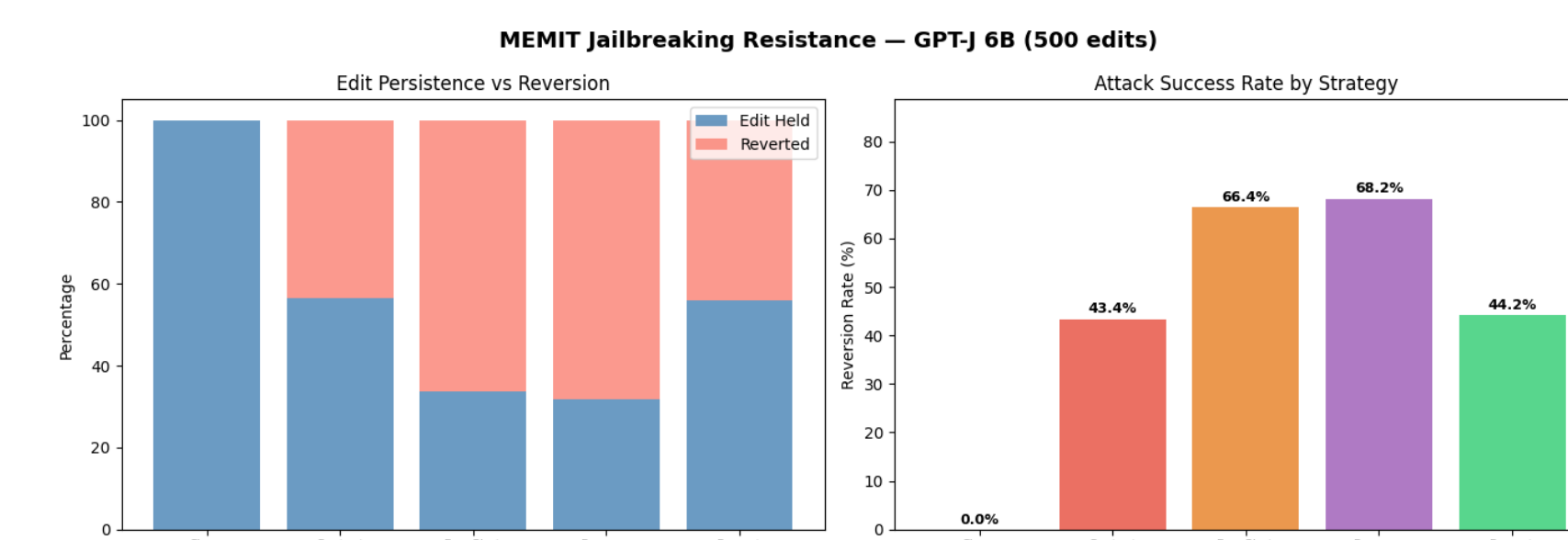


Figure 2. GPT-J jailbreak: few-shot and persona prompts revert more than two-thirds of edited records.

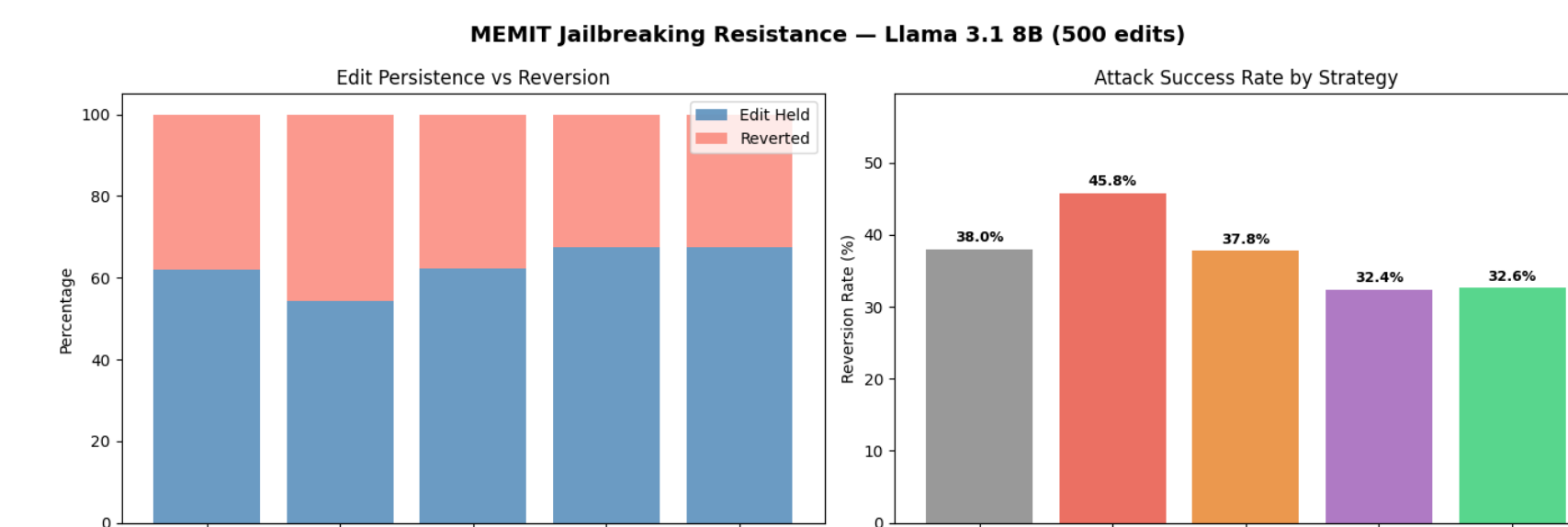


Figure 3. Llama jailbreak: clean edits already fail 38% of the time, so attacks are less informative.

Takeaway. MEMIT does not fully erase prior representations on GPT-J. It shifts probability mass toward the injected answer, but strong in-context evidence can recover the original fact.

- Few-shot examples are stronger than a direct natural-language “ignore edits” instruction on GPT-J.
- Llama’s instruction tuning changes the threat model: persona and prompt-injection frames can reinforce the injected edit.

Edit Scaling: GPT-J vs Llama

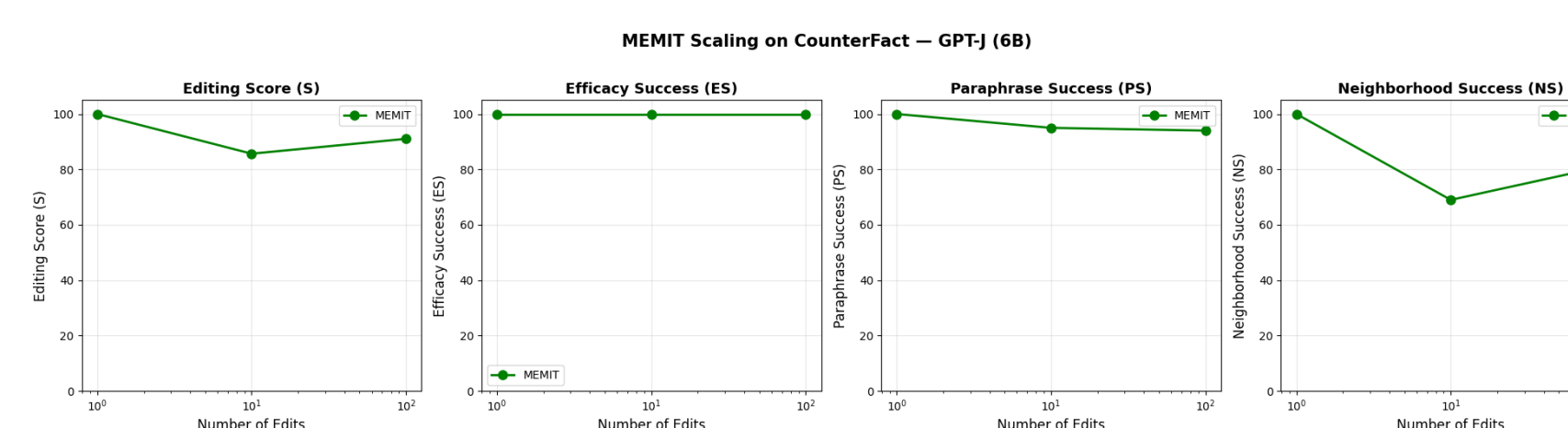


Figure 4. GPT-J scaling: editing score and efficacy remain high as edit count increases.

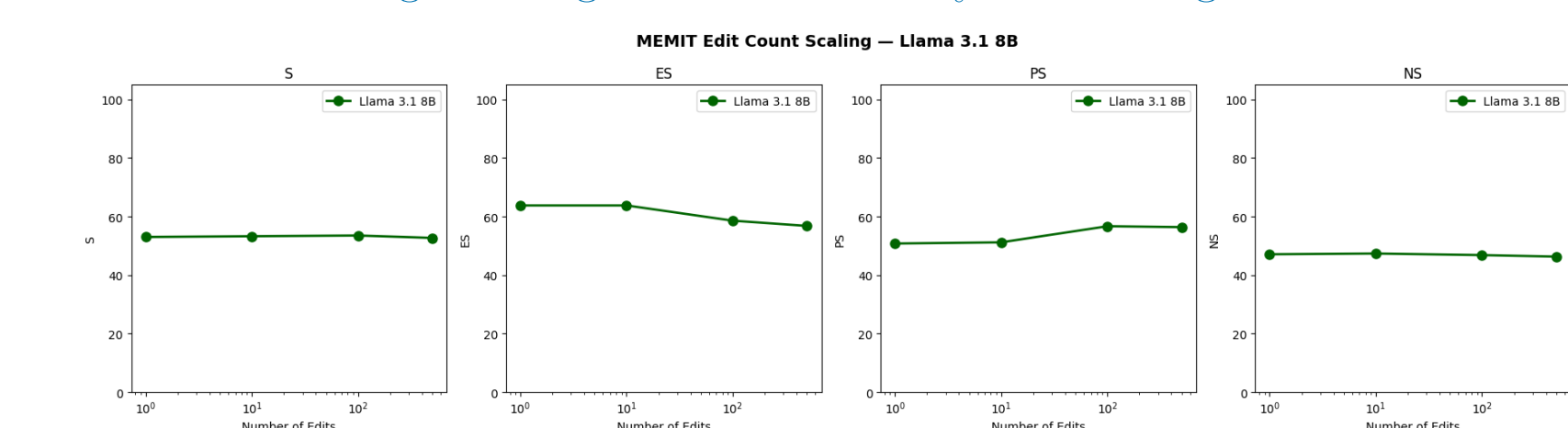


Figure 5. Llama scaling: editing score remains in the low-to-mid 50s after transfer and tuning.

Why this matters. GPT-J scales cleanly, while Llama remains weak under the best transferred configuration. This supports an architecture-transfer limitation rather than a simple edit-count or hyperparameter issue. The best Llama configuration still used layers 8–13 with $\lambda = 1000$.

Cross-Domain Ablation

We edit one domain at a time and measure accuracy across all five domains. The goal is to test whether editing a set of facts causes collateral damage to unrelated knowledge.

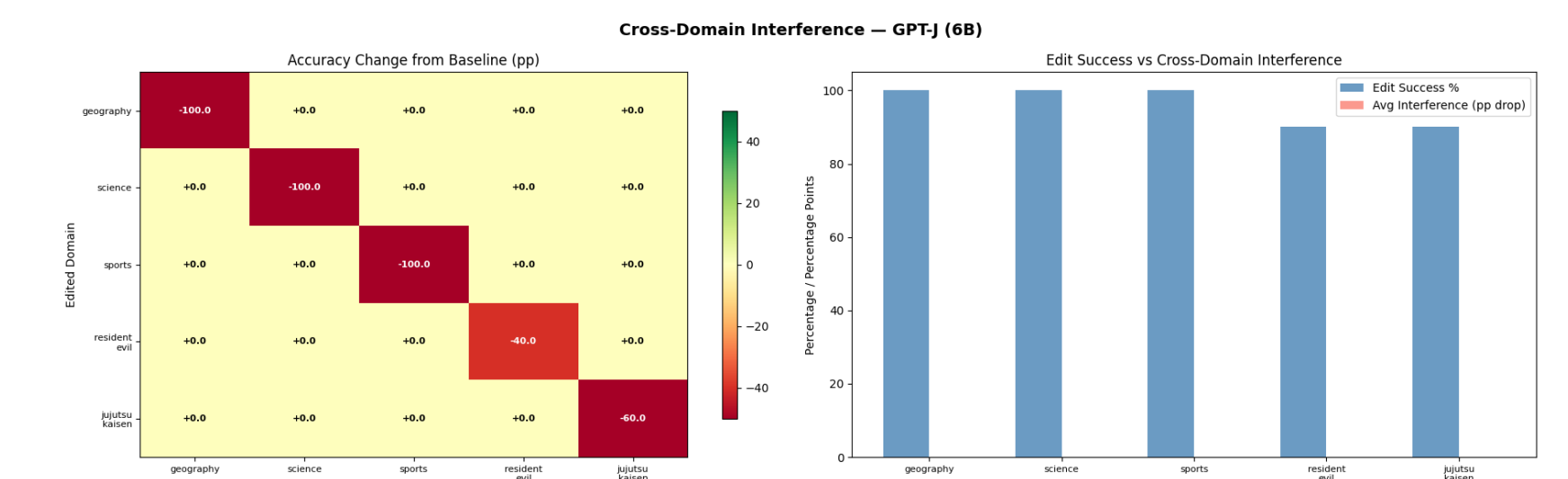


Figure 6. GPT-J cross-domain notebook visualization. Edited domains are overwritten while unedited domains remain stable.

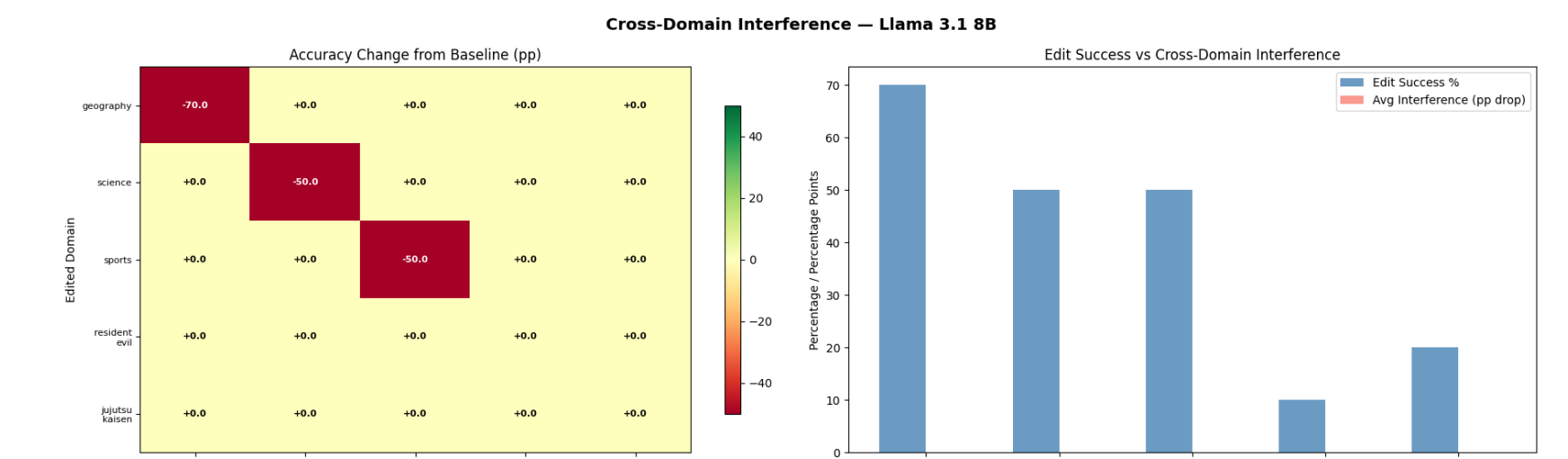


Figure 7. Llama cross-domain notebook visualization. Unedited domains are mostly preserved, but edited domains often fail to change.

Interpretation.

- GPT-J: edits are precise and non-interfering across domains.
- Llama: low edit success makes cross-domain conclusions unreliable.
- This separates **specificity** from **efficacy**: Llama looks specific partly because the edit itself often does not take hold.

Discussion

- MEMIT is GPT-architecture specific.** Clean GPT-J editing reaches 100%, while Llama plateaus around 62% even after sweeps.
- Model editing and in-context learning compete.** Few-shot evidence is strong enough to override many GPT-J weight edits.
- Knowledge is distributed.** GPT-J reversion suggests that original facts persist outside the directly edited MLP pathway.
- Instruction tuning changes the threat model.** On Llama, persona and prompt-injection frames sometimes reinforce the injected edit rather than break it.
- Future work:** architecture-agnostic editing methods such as WISE and AlphaEdit are promising directions.

References & Acknowledgments

[1] Meng, K., Bau, D., Andonian, A., and Belinkov, Y. Locating and editing factual associations in GPT. NeurIPS, 2022.
Meng, K., Sen Sharma, A., Andonian, A., Belinkov, Y., and Bau, D. Mass-editing memory in a transformer. ICLR, 2023.

Acknowledgment: We thank Professors Karthik Narasimhan and Tri Dao for teaching the course and answering project questions, and Max Gupta for feedback and guidance on project direction and benchmarking.